	Network Administration ***Installing and Configuring a DHCP***	2025/2026

Objectifs :
• Understand and implement DHCP services • Verify and troubleshoot DHCP operation

Background / Scenario

The DHCP server is essential for automatically managing the allocation of IP addresses in a network. This process is important for simplifying network management and optimizing connectivity between different devices. The server has three methods of assigning an IP address:

- ***Manual assignment***: the administrator associates an IP address with a MAC address.
- ***Dynamic allocation***: the server supplies the address from a set of reserved addresses, but possession of the address is limited in time.

This Lab focuses on Dynamic allocation, which allows a host, unknown to the network administrator, to be assigned a new IP address from a set of addresses for that network. To do this, the network administrator allocates a set of addresses in each subnet.

During startup, the client station, which lacks an assigned IP address, initiates a "DHCP DISCOVER" broadcast message to request an IP address. This message is broadcasted to the address 255.255.255.255, which serves as the broadcast address. Both machines on the network and the DHCP server(s) receive this request.

Since the network creation and routing are already done in Lab0, this lab's objective focuses only on DHCP concepts and actions. More precisely,

- Configure and operate a DHCP server capable of dynamically assigning IP addresses and essential network parameters to multiple client hosts across distinct subnets.
- To analyze and validate the DHCP address allocation mechanism by verifying automatic IP configuration on client systems and ensuring the correctness of assigned network parameters, such as address ranges, default gateways, and lease durations.

This lab considers that the laboratory environment is composed of two local area networks (LANs) and one wide area network (WAN), as in previous Lab0:

- LAN 1:
  - PC1: 192.168.10.10/24 (assigned statically)
  - Gateway (LAN1): 192.168.10.1/24
- LAN 2

- PC2: 192.168.20.10/24 (assigned statically)
- Gateway (LAN2): 192.168.20.1/24

- WAN
  - ISP: 209.165.200.225/27
  - Gateway (WAN interface): 209.165.200.226/27

In this lab, we consider that the gateway is statically configured, while PC1 and PC2 obtain their IP configuration dynamically using DHCP.

Part I: Dynamic Allocation

In this Lab, follow these steps for deployment and testing. A dedicated namespace is used to represent the DHCP server, ensuring isolation from clients and gateways. The DHCP server is connected to LAN 1 via interface dhcp1 and LAN 2 via interface dhcp2.

Each interface is assigned a static IP address. These addresses correspond to the default gateways distributed to clients in each LAN.

- dhcp1: 192.168.10.1/24
- dhcp2: 192.168.20.1/24

Step 1: Preparation and Environment Cleanup[1]

This step ensures that the DHCP server namespace and its associated virtual Ethernet interfaces are fully removed. It guarantees a clean environment for reconfiguring DHCP without affecting the existing client or gateway namespaces.

```
# Stop any running DHCP service
sudo ip netns exec DHCP_SERVER pkill dnsmasq 2>/dev/null || true
# Remove DHCP configuration and lease files (inside namespace if exists, otherwise on host)
sudo ip netns exec DHCP_SERVER rm -f /tmp/dhcp_lan1.conf /tmp/dhcp_lan2.conf /var/lib/misc/dnsmasq.leases 2>/dev/null || true
sudo rm -f /tmp/dhcp_lan1.conf /tmp/dhcp_lan2.conf /var/lib/misc/dnsmasq.leases 2>/dev/null
# Delete leftover veth interfaces on the host (if they exist)
sudo ip link delete dhcp_pc1 2>/dev/null || true
sudo ip link delete dhcp_pc2 2>/dev/null || true
# Delete old DHCP_SERVER namespace (optional, ensures fully clean start)
sudo ip netns del DHCP_SERVER 2>/dev/null || true
```

[1] Note: Do not delete PC1, PC2, GATEWAY, or ISP namespaces, we are only cleaning DHCP-related components.

Step 2: Creation of the DHCP Server Namespace

```
# This isolation allows the DHCP server to operate independently, closely simulating a real network deployment.
# Only create if it does not exist
sudo ip netns add DHCP_SERVER 2>/dev/null || echo "DHCP_SERVER namespace exists, reusing"
```

Step 3: Creation of Virtual Ethernet Interfaces

```
# Create veth pairs to connect DHCP server to each LAN
sudo ip link add dhcp_pc1 type veth peer name dhcp1
sudo ip link add dhcp_pc2 type veth peer name dhcp2
# Assign DHCP-side interfaces to the DHCP_SERVER namespace
sudo ip link set dhcp1 netns DHCP_SERVER 2>/dev/null || echo "dhcp1 already in DHCP_SERVER"
sudo ip link set dhcp2 netns DHCP_SERVER 2>/dev/null || echo "dhcp2 already in DHCP_SERVER"
# Assign client-side interfaces to PC1 and PC2 namespaces
sudo ip link set dhcp_pc1 netns PC1 2>/dev/null || echo "dhcp_pc1 already in PC1"
sudo ip link set dhcp_pc2 netns PC2 2>/dev/null || echo "dhcp_pc2 already in PC2"
# Bring interfaces up
sudo ip netns exec DHCP_SERVER ip link set dhcp1 up
sudo ip netns exec DHCP_SERVER ip link set dhcp2 up
sudo ip netns exec DHCP_SERVER ip link set lo up
# Bring interfaces up
sudo ip netns exec PC1 ip link set dhcp_pc1 up
sudo ip netns exec PC2 ip link set dhcp_pc2 up
```

Step 4: IP Address Configuration on the DHCP Server

```
# Each DHCP server interface is assigned a static IP address corresponding to the gateway address of its LAN. Later distributed to clients as default gateway options via DHCP.
sudo ip netns exec DHCP_SERVER ip addr add 192.168.10.1/24 dev dhcp1
sudo ip netns exec DHCP_SERVER ip addr add 192.168.20.1/24 dev dhcp2
# The interfaces are then activated
sudo ip netns exec DHCP_SERVER ip link set dhcp1 up
sudo ip netns exec DHCP_SERVER ip link set dhcp2 up
sudo ip netns exec DHCP_SERVER ip link set lo up
```

Step 5; Installation of the DHCP Service (dnsmasq)

```
# The DHCP service is implemented using dnsmasq. This ensures that the DHCP daemon is available inside the DHCP server namespace.
sudo ip netns exec DHCP_SERVER bash -c "command -v dnsmasq >/dev/null || apt update && apt install -y dnsmasq"
# After running the command, you can confirm that dnsmasq is installed inside the namespace.
sudo ip netns exec DHCP_SERVER dnsmasq --version
```

Step 6: DHCP Configuration for LAN 1

 # A DHCP configuration file is created for LAN 1. This configuration:

- *# Restricts DHCP service to interface dhcp1*
- *# Defines an IP pool within 192.168.10.0/24*
- *# Specifies the default gateway as 192.168.10.1*
- *# Sets a lease duration of 12 hours*

```
sudo ip netns exec DHCP_SERVER bash -c 'cat > /tmp/dhcp_lan1.conf <<EOF
interface=dhcp1
dhcp-range=192.168.10.50,192.168.10.100,12h
dhcp-option=3,192.168.10.1
EOF'
```

Step 7: DHCP Configuration for LAN 2

```
# A second configuration file is created for LAN 2. This allows the same DHCP server to manage multiple subnets independently.
sudo ip netns exec DHCP_SERVER bash -c 'cat > /tmp/dhcp_lan2.conf <<EOF
interface=dhcp2
dhcp-range=192.168.20.50,192.168.20.100,12h
dhcp-option=3,192.168.20.1
EOF'
```

Step 8; Launching the DHCP Service

```
# Any existing dnsmasq process is terminated to avoid conflicts. This enables concurrent DHCP service on both LANs.
sudo ip netns exec DHCP_SERVER pkill dnsmasq || true
# The DHCP service is then started with both configuration files
sudo ip netns exec DHCP_SERVER dnsmasq \
--conf-file=/tmp/dhcp_lan1.conf \
--conf-file=/tmp/dhcp_lan2.conf \
--conf-dir= --no-daemon &
# Check if running
sudo ip netns exec DHCP_SERVER ps aux | grep [d]nsmasq
```

Step 9: Enabling DHCP Clients

```
# The client interfaces connected to the DHCP server are activated.
sudo ip netns exec PC1 ip link set dhcp_pc1 up
sudo ip netns exec PC2 ip link set dhcp_pc2 up
```

Step 10: Enabling DHCP Clients

```
# The DHCP client (dhclient) is executed on each host. Initiate the DHCP discovery and lease acquisition process.
#On PC1 mini-shell            # On PC2 minishell
dhclient -v dhcp_pc1          dhclient -v dhcp_pc2
```

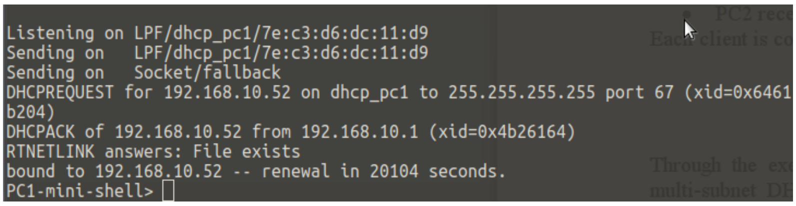
Step 11: Verification of DHCP Address Allocation

```
# The assigned IP addresses are verified using:
# On PC1
ip addr show dhcp_pc1
cat /var/lib/dhcp/dhclient.leases
ping -c 3 192.168.10.1
On PC2
ip addr show dhcp_pc2
cat /var/lib/dhcp/dhclient.leases
ping -c 3 192.168.20.1
```

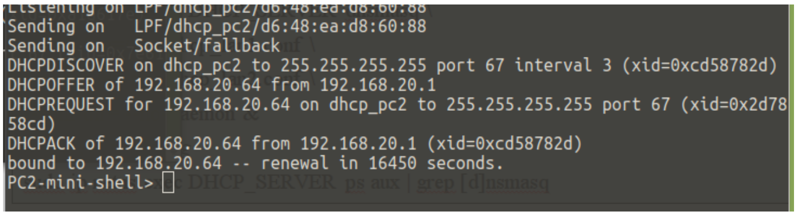
Successful output confirms that:

- PC1 receives an address in 192.168.10.0/24: 192.168.10.52
- PC2 receives an address in 192.168.20.0/24: 192.168.20.64

Each client is correctly configured with its respective gateway



```
Listening on LPF/dhcp_pc1/7e:c3:d6:dc:11:d9
Sending on   LPF/dhcp_pc1/7e:c3:d6:dc:11:d9
Sending on   Socket/fallback
DHCPREQUEST for 192.168.10.52 on dhcp_pc1 to 255.255.255.255 port 67 (xid=0x6461
b204)
DHCPACK of 192.168.10.52 from 192.168.10.1 (xid=0x4b26164)
RTNETLINK answers: File exists
bound to 192.168.10.52 -- renewal in 20104 seconds.
PC1-mini-shell>
```



```
Listening on LPF/dhcp_pc2/d6:48:ea:d8:60:88
Sending on   LPF/dhcp_pc2/d6:48:ea:d8:60:88
Sending on   Socket/fallback
DHCPDISCOVER on dhcp_pc2 to 255.255.255.255 port 67 interval 3 (xid=0xcd58782d)
DHCPOFFER of 192.168.20.64 from 192.168.20.1
DHCPREQUEST for 192.168.20.64 on dhcp_pc2 to 255.255.255.255 port 67 (xid=0x2d78
58cd)
DHCPACK of 192.168.20.64 from 192.168.20.1 (xid=0xcd58782d)
bound to 192.168.20.64 -- renewal in 16450 seconds.
PC2-mini-shell>
```

Step 12: Bash script to automate the DHCP lab

Create a Bash script to automate the DHCP lab setup so that all steps run in a single script instead of executing commands step by step.

Step 13: Bash script to clean and reset the DHCP lab

Create a Bash script to clean and reset the DHCP lab environment so that the DHCP setup could be safely re-run without rebuilding the entire virtual network (Lab 0). The cleanup script must automatically perform the following:

- Stop the DHCP service (dnsmasq) running inside the DHCP_SERVER namespace
- Remove all DHCP configuration files and lease files
- Delete DHCP-related virtual interfaces created for the overlay (dhcp_pc1, dhcp_pc2)
- Remove the DHCP_SERVER network namespace if it exists
- Preserve the original Lab 0 topology (PC1, PC2, GATEWAY, ISP remain untouched)

The goal of this script is to restore the network to its pre-DHCP state, allowing:

- Static IP configuration from Lab 0 to continue working
- The DHCP lab to be re-executed multiple times without conflicts

Step 14: Dynamic NAT script (Lab 3)

After DHCP is set up, NAT (**Network Address Translation**) can be executed and it will enable dynamic NAT for both LAN1 and LAN2. After running, PC1 and PC2 should be able to reach the ISP namespace. The NAT process can be launched, only if these conditions are met. **See Lab3 for more details**.

```
#DHCP clients already have IPs from the server (check ip addr show dhcp_pc1 / dhcp_pc2).
sudo ip netns exec PC1 ip addr show dhcp_pc1
sudo ip netns exec PC2 ip addr show dhcp_pc2
# Gateway namespace is configured with LAN and WAN interfaces.
sudo ip netns exec GATEWAY ip addr show
# IP forwarding is enabled on the gateway
sudo ip netns exec GATEWAY sysctl -w net.ipv4.ip_forward=1

#Test connectivity from PC1 and PC2 can be checked
sudo ip netns exec PC1 ping -c 3 209.165.200.225
sudo ip netns exec PC2 ping -c 3 209.165.200.225
```

Deliverable (To be uploaded on the Platform

Through the execution of these commands, the lab demonstrates the deployment of a multi-subnet DHCP server using dnsmasq, the automatic configuration of client hosts, and the practical operation of DHCP within an isolated Linux network namespace environment. The main report document (word/pdf, 10 pages Max) must include the following:

- Introduction / Background
  - Brief explanation of DHCP, static/dynamic IP assignment, network namespaces, and veth interfaces.
- Network Topology
  - Diagram showing namespaces (PC1, PC2, DHCP_SERVER, GATEWAY), veth links, and IP addressing.
- DHCP Configuration
  - Server setup (dnsmasq), LAN interfaces, IP ranges, default gateways.
- Procedure / Methodology
  - Step-by-step setup (cleanup, namespaces, veths, IPs, DHCP server, client configuration).
- Verification and Testing
  - IP addresses assigned to PC1 and PC2. DHCP lease files. Ping tests to gateways.
- Results
  - Tables or screenshots of assigned IPs, lease times, and connectivity tests.
- Conclusion
  - Summary of achieved objectives.

A single compressed file named in the format “lastname.firstname-Lab2.zip”, must be uploaded on the platform (see deadlines) and must include:

- The main report document (Word or PDF)
- All developed Bash scripts

Appendix

Network namespace basics (FOUNDATION)

```
ip netns list
ip netns exec <ns> <command>
```

Interface inspection

```
ip addr show <iface>
# Exemple
sudo ip netns exec PC1 ip a
sudo ip netns exec PC2 ip addr show dhcp_pc2
```

Routing verification

```
ip route show
# Exemple
sudo ip netns exec PC1 ip route
```

DHCP client control

```
# Release a lease
dhclient -r <interface>
# Request a lease
dhclient <interface>
# Exemple
sudo ip netns exec PC1 dhclient -r dhcp_pc1
sudo ip netns exec PC1 dhclient dhcp_pc1
```

DHCP client control

```
# Release a lease
dhclient -r &lt;interface&gt;
# Request a lease
dhclient &lt;interface&gt;
# Exemple
sudo ip netns exec PC1 dhclient -r dhcp_pc1
sudo ip netns exec PC1 dhclient dhcp_pc1
```

DHCP client control

```
# Start DHCP server
dnsmasq --conf-file=<file> --no-daemon
# Exemple
sudo ip netns exec DHCP_SERVER dnsmasq \
  --conf-file=/tmp/dhcp_lan1.conf \
  --conf-file=/tmp/dhcp_lan2.conf
# Stop DHCP server
pkill dnsmasq
```

Observing DHCP results

```
# DHCP configuration files
hostname -I
ip addr show
# Exemple
sudo ip netns exec PC1 hostname –I
#  Packet capture (DHCP exchange)
tcpdump -i dhcp_pc1 -n port 67 or 68
# Exemple, DHCP uses: UDP 67 (server) and UDP 68 (client)
sudo ip netns exec PC1 tcpdump -i dhcp_pc1 -n port 67 or 68
# See leases
cat /var/lib/misc/dnsmasq.leases
# Exemple
sudo ip netns exec DHCP_SERVER cat /var/lib/misc/dnsmasq.leases
```